Hypothesis

HW 3: Model-based Bias Removal & Physiological Curve Fitting

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Part A: Modeling blood pressure as a function of age:

- To fit age-stratified SBP(systolic) and DBP (diastolic) blood pressure data for males and females using two models (might experiment with more later):

1) Quadratic polynomial models:

$SBP(a) = c_1a^2 + c_2a + c_3$ & $DBP(a) = d_1a^2 + d_2a + d_3$

2) Biological non-linear models:

- Sigmoids function for SBP (helps capture plateaus).
- Gaussian function for DBP (to capture possible mid life peaks).

- Planning to use linear and non-linear least square regression.

Evaluation:

- MSE (Mean Squared Error), R^2 (coefficient of determination) and AIC/BIC to maintain a balance between goodness of fit and complexity of parameters.

(I expect the sigmoid model for SBP and gaussian model for DBP to outperform polynomials. I also expect the AIC/BIC to heavily penalize the extra parameters of non-linear models (sig & gaussian) unless SOS errors are reduced substantially).

Part B: Model-based bias removal using Synthetic data.

- generate 60,000 samples using bivariate normal distributions.
- Construct datasets across four imbalance ratios (50/50, 75/25, 90/10, 97/3). Assign labels, scatter plots to visualize feature distributions
- 70:30 train-test split
- train binary classifier (log. reg. or SVM) to predict gender.
- Evaluate using ROC curves/AUC, accuracy, F1, per-sex recall.
- incorporate bias-penalized loss function $c(\theta) = f(\theta) + \lambda\phi(\theta) \rightarrow \Delta$recall
- sweep the penalty or adjust decision thresholds via grid search to generate and plot Accuracy-Equity tradeoff curve.
- augment skewed 90/10 training set with synthetic minority (F) via 2 approaches: samples directly from true bivariate distribution, and (b) sampled uniformly.
- Comparing performance.

(I think as imbalance reaches 97/3, overall accuracy will be crazy high while female recall is down to zero. The 90/10 model will probably misclassify female profiles as male due to skewed boundaries, but the penalized loss and augmentation are expected to restore female recall, even though it reduces overall accuracy.